

- [3pts] 1) Suppose that  $x > 0$  and  $y < 0$ . Using the definition of absolute value and splitting into cases, show that  $|x + y| \leq |x - y|$ .

*Solution:*

Recall the definition of absolute values as follows:

$$(a) \ z = \begin{cases} z & \text{for } z \geq 0 \\ -z & \text{for } z < 0 \end{cases}$$

Suppose  $x > 0$  and  $y < 0$ , then to show that the inequality holds, it suffices to consider the following cases:

- i)  $(x + y) \geq 0$  and  $(x - y) \geq 0$ :

From the above assumptions and with  $x > 0$  and  $y < 0$ , then

$$\begin{aligned} |x + y| \geq 0 &\implies (x + y) \geq 0 \\ |x - y| \geq 0 &\implies (x - y) \geq 0 \\ &\implies 0 \leq (x - y) - (x + y) \\ &\quad = |x - y| - |x + y| \\ &\implies |x + y| \leq |x - y| \end{aligned}$$

- ii)  $|x + y| < 0$  and  $|x - y| < 0$ :

From the above assumptions and with  $x > 0$ ,  $y < 0$  and  $x \neq y$ , then

$$\begin{aligned} |x + y| < 0 &\implies -(x + y) < 0 \\ |x - y| < 0 &\implies -(x - y) < 0 \\ &\implies (x + y) > 0 \text{ and } (x - y) > 0 \\ &\implies 0 < (x - y) - (x + y) \\ &\quad = |x - y| - |x + y| \\ &\implies |x + y| \leq |x - y| \end{aligned}$$

- iii)  $|x + y| < 0$  and  $|x - y| \geq 0$  (or vice versa):

Without loss of generality, assume the above assumptions and with  $x > 0$ ,  $y < 0$  and  $x \neq y$ , then

$$\begin{aligned} |x + y| < 0 &\implies -(x + y) < 0 \\ &\implies (x + y) > 0 \\ |x - y| \geq 0 &\implies (x - y) \geq 0 \\ &\implies 0 \leq (x + y) + (x - y) \\ &\quad = (x + y) + |x - y| \\ -(x + y) &\leq |x - y| \\ |x + y| &\leq |x - y| \end{aligned}$$

For the equality to hold in the given expression, the given conditions may not suffice. That is,

$$|x + y| = |x - y|$$

when  $x \geq 0$  **and**  $y \geq 0$ , or when  $x \leq 0$  **and**  $y \leq 0$ , or essentially when  $x$  and  $y$  have the same parity.  $\square$

### 3.4: Quadratic Equations and Problem Solving

[3pts] 23c) Find the order of convergence of the following sequence:

$$x_n = \left(1 + \frac{1}{n}\right)^{\frac{1}{2}}$$

*Solution:*

To find the order of convergence of the given series, observe the following:

$$\begin{aligned}x_n &= \left(1 + \frac{1}{n}\right)^{\frac{1}{2}} \implies x_n^2 = 1 + \frac{1}{n} \\n &= \frac{1}{x_n^2 - 1} \\n + 1 &= \frac{x_n^2}{x_n^2 - 1}\end{aligned}$$

Thus,

$$x_{n+1} = \left(1 + \frac{(x_n^2 - 1)}{x_n^2}\right)^{\frac{1}{2}} = \left(\frac{2x_n^2 - 1}{x_n^2}\right)^{\frac{1}{2}}$$

Hence,

$$F(x_n) = \frac{\sqrt{2x_n^2 - 1}}{x_n} \implies F'(x_n) = \frac{1}{x_n^2 \sqrt{2x_n^2 - 1}}.$$

At  $\xi = 1$ ,  $F'(\xi) \neq 0$  and the convergence is a linear order.

□

- [4pts] 32) Let  $\frac{1}{2} \leq q \leq 1$  and define  $F(x) = 2x - qx^2$ . On what interval can it be guaranteed that the method of iteration using  $F$  will converge to a fixed point?

*Solution:*

Recall from the Contractive Mapping Theorem: if  $F(x) = 2x - qx^2$  is a contractive mapping, then  $x_{n+1} = F(x_n)$  will converge to a unique fixed point  $s$ . Thus, to determine the interval to guarantee that the function will converge to a fixed point, it suffices to find some interval such that the definition of a contractive function holds

$$|F'(x)| < 1$$

for all  $x \in C$ .

Given  $F(x) = 2x - qx^2$  with  $\frac{1}{2} \leq q \leq 1$ , differentiate to find the critical value(s) to determine a potential maximum or minimum bound:

$$\begin{aligned} F'(x) &= 2 - 2qx \\ 0 &= 2 - 2qx \implies x = \frac{1}{q} \\ |F'(x)| \leq 1 &\implies |2 - 2qx| \leq 1 \\ &\iff -1 < 2(1 - qx) < 1 \\ &\iff -\frac{1}{2} < 1 - qx < \frac{1}{2} \\ &\iff \frac{3}{2} > qx > \frac{1}{2} \\ &\iff \frac{1}{2} < qx < \frac{3}{2} \\ &\implies \frac{1}{2q} < x < \frac{3}{2q} \end{aligned}$$

From the found critical point,  $x = \frac{1}{q}$ , we can determine the bound of  $F(x)$  as follows:

$$\begin{aligned} F\left(\frac{1}{2q}\right) &= \frac{1}{q} - \frac{1}{4q} = \frac{3}{4q} \\ F\left(\frac{1}{q}\right) &= \frac{2}{q} - \frac{1}{q} = \frac{1}{q} \\ F\left(\frac{3}{2q}\right) &= \frac{3}{q} - \frac{9}{4q} = \frac{3}{4q} \end{aligned}$$

Since  $\frac{1}{2} \leq q \leq 1$ , then the above yields

$$\frac{3}{4q} \leq F(x) \leq \frac{1}{q}$$

for all  $\frac{1}{2q} < x < \frac{3}{2q}$ . Thus, on  $\left(\frac{1}{2q}, \frac{3}{2q}\right)$ , it can be guaranteed that  $F(x)$  will converge to a fixed point.

□